

# Analytic Geometry

## Marking Scheme & Blueprint

MTH1W · Module 6 Test — Teacher Copy

### Blueprint

Curriculum: MTH1W (Grade 9 de-streamed math). Total: 70 marks · 60 minutes · 30 questions. Categories: K/U, Application (App), Thinking (Think), Communication (Comm).

| Q   | Sub-topic                | Level | Marks | Category | Section |
|-----|--------------------------|-------|-------|----------|---------|
| A1  | Slope                    | L1    | 1     | K/U      | A       |
| A2  | Slope                    | L1    | 1     | K/U      | A       |
| A3  | Graphing                 | L1    | 1     | K/U      | A       |
| A4  | Parallel                 | L1    | 1     | K/U      | A       |
| A5  | Perpendicular            | L1    | 1     | K/U      | A       |
| A6  | Midpoint                 | L1    | 1     | K/U      | A       |
| A7  | Length                   | L1    | 1     | K/U      | A       |
| A8  | Slope                    | L1    | 1     | K/U      | A       |
| A9  | Graphing                 | L1    | 1     | K/U      | A       |
| A10 | Slope                    | L1    | 2     | K/U      | A       |
| A11 | Length                   | L1    | 2     | K/U      | A       |
| A12 | Midpoint                 | L1    | 2     | K/U      | A       |
| A13 | Perpendicular            | L1    | 2     | K/U      | A       |
| A14 | Parallel                 | L1    | 2     | K/U      | A       |
| A15 | Graphing                 | L1    | 2     | K/U      | A       |
| B16 | Slope                    | L2    | 2     | App      | B       |
| B17 | Length                   | L2    | 2     | App      | B       |
| B18 | Midpoint                 | L2    | 2     | App      | B       |
| B19 | Parallel                 | L2    | 2     | App      | B       |
| B20 | Perpendicular            | L2    | 2     | App      | B       |
| B21 | Graphing                 | L2    | 2     | App      | B       |
| B22 | Slope                    | L2    | 3     | App      | B       |
| B23 | Graphing                 | L2    | 3     | App      | B       |
| B24 | Length                   | L2    | 3     | App      | B       |
| C25 | Perpendicular            | L3    | 4     | Think    | C       |
| C26 | Parallel                 | L3    | 4     | Think    | C       |
| C27 | Midpoint & Length        | L3    | 5     | Think    | C       |
| C28 | Perpendicular & Graphing | L3    | 5     | Think    | C       |
| C29 | Length & Right Triangle  | L3    | 6     | Think    | C       |
| D30 | Parallel Lines Reasoning | L3    | 4     | Comm     | D       |

**By category (marks):** K/U 21 · Application 21 · Thinking 24 · Communication 4

**By level (marks):** L1 21 · L2 21 · L3 28 → L1 30% · L2 30% · L3 40%

## Worked Solutions & Mark Allocation

- 1. (A1 · K/U · 1 marks)**  
 $m = 7$ . (A1)
- 2. (A2 · K/U · 1 marks)**  
 $m = -1$ . (A1)
- 3. (A3 · K/U · 1 marks)**  
(0, -8). (A1)
- 4. (A4 · K/U · 1 marks)**  
 $m = 6$ . (A1)
- 5. (A5 · K/U · 1 marks)**  
 $-1/2$ . (A1)
- 6. (A6 · K/U · 1 marks)**  
(2, 5). (A1)
- 7. (A7 · K/U · 1 marks)**  
7. (A1)
- 8. (A8 · K/U · 1 marks)**  
 $9/3 = 3$ . (A1)
- 9. (A9 · K/U · 1 marks)**  
Yes; it is the y-intercept. (A1)
- 10. (A10 · K/U · 2 marks)**  
 $(14 - 2)/(5 - 1) = 12/4 = 3$ . (M1, A1)
- 11. (A11 · K/U · 2 marks)**  
 $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$ . (M1, A1)
- 12. (A12 · K/U · 2 marks)**  
 $((-2+6)/2, (4+10)/2) = (2, 7)$ . (M1, A1)
- 13. (A13 · K/U · 2 marks)**  
Given slope -3; perpendicular  $1/3$ . (M1, A1)
- 14. (A14 · K/U · 2 marks)**  
 $y = 2x + 5$ . (M1, A1)
- 15. (A15 · K/U · 2 marks)**  
 $0 = -x + 4 \rightarrow x = 4 \rightarrow (4, 0)$ . (M1, A1)
- 16. (B16 · Application · 2 marks)**  
 $m = (-7 - 5)/(4 - (-2)) = -12/6 = -2$ . (M1, A1)

- 17. (B17 • Application • 2 marks)**  
 $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ . (M1, A1)
- 18. (B18 • Application • 2 marks)**  
 $((-5+7)/2, (-2+8)/2) = (1, 3)$ . (M1, A1)
- 19. (B19 • Application • 2 marks)**  
 Both have slope 4; equal slopes, so yes — parallel. (M1, A1)
- 20. (B20 • Application • 2 marks)**  
 Negative reciprocal of  $3/4$  is  $-4/3$ . (M1, A1)
- 21. (B21 • Application • 2 marks)**  
 y-intercept (0, -12); x-intercept:  $0 = 3x - 12 \rightarrow x = 4 \rightarrow (4, 0)$ . (M1, A1)
- 22. (B22 • Application • 3 marks)**  
 $(k - 3)/(8 - 2) = 2$ , so  $k - 3 = 12$ . (M1, M1)  
 $k = 15$ . (A1)
- 23. (B23 • Application • 3 marks)**  
 $y = -2x + 6$ . (M1, A1)  
 Slope = -2; y-intercept (0, 6). (M1)
- 24. (B24 • Application • 3 marks)**  
 $AB = 9$ ,  $BC = 12$ ,  $CA = \sqrt{9^2 + 12^2} = \sqrt{225} = 15$ . (M1, M1)  
 Perimeter =  $9 + 12 + 15 = 36$ . (A1)
- 25. (C25 • Thinking • 4 marks)**  
 Slope =  $(8 - 2)/(4 - 1) = 6/3 = 2$ . (M1, A1)  
 Perpendicular slope = negative reciprocal of 2 =  $-1/2$ . (M1, A1)
- 26. (C26 • Thinking • 4 marks)**  
 Slope =  $(5 - (-3))/(2 - 0) = 8/2 = 4$ ; y-intercept (0, -3), so  $y = 4x - 3$ . (M1, A1)  
 Same slope as  $y = 4x + 7$ , so yes — parallel. (M1, A1)
- 27. (C27 • Thinking • 5 marks)**  
 $B = (2 \cdot 4 - 1, 2 \cdot 3 - (-1)) = (7, 7)$ . (M1, A1)  
 $AB = \sqrt{(7-1)^2 + (7-(-1))^2} = \sqrt{36 + 64} = \sqrt{100}$ . (M1)  
 $= 10$ . (A1, A1)
- 28. (C28 • Thinking • 5 marks)**  
 Perpendicular slope = negative reciprocal of  $1/2 = -2$ ; y-intercept 5, so  $y = -2x + 5$ .  
 (M1, A1)  
 x-intercept:  $0 = -2x + 5 \rightarrow x = 5/2$ . (M1)  
 x-intercept (2.5, 0). (A1, A1)

**29. (C29 • Thinking • 6 marks)**

PQ = 6 (horizontal), QR = 8 (vertical). (M1, M1)

PR =  $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ . (M1)

Perimeter =  $6 + 8 + 10 = 24$ . (A1, A1)

Since  $6^2 + 8^2 = 36 + 64 = 100 = 10^2$ , the angle at Q is  $90^\circ$ . (A1)

**30. (D30 • Communication • 4 marks)**

Both lines have slope 2, so they are equally steep (parallel). (A1)

Their y-intercepts differ (1 and 6), so they are not the same line. (A1)

Distinct parallel lines never intersect, so the claim is false. (C1, C1)